Bilateral Adaptive Evolution Transformer for Multispectral Image Fusion

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***Abstract*— Pansharpening is the critical technology for generating high-resolution (HR) multispectral (MS) images by learning the cross-modality complementary representations between the panchromatic (PAN) images and low-resolution (LR) MS images. Though methods based on convolutional neural networks (CNNs) have dominated the pansharpening community, they still suffer from the limited global modeling capability due to the inherent property of the convolutional operator. To remedy this common limitation, the transformer family has recently gained great popularity in this field. However, existing cascaded transformer designs inevitably introduce a heavy memory footprint and computational cost due to the dense dot-product self-attention (SA) computation. More importantly, these paradigms simply ignore the innate sparsity of remote sensing images, leading to information redundancy and a challenging optimization process. To alleviate these issues, we propose the bilateral adaptive evolution transformer (BAEFormer), which is built upon two core mechanisms: bilateral attention computation and adaptive attention evolution. Specifically, we first decompose the conventional quadratic complexity SA into linear-degree height and width computing at the first stage, respectively, which significantly reduces the computational complexity. Given the data-specific properties, furthermore, we devise a novel yet effective neighboring layer-dependent strategy to adaptively update the attention map of two spatial dimensions, thereby avoiding the repetitive SA computation while taking into account the dynamics toward the evolution of attention weights. Our model, called BAEFormer, outperforms other state-of-the-art pansharpening methods on various remote sensing datasets while showing fewer network parameters and computational requirements. The code is available at https://github.com/coder-JMHou/BAEFormer.**

***Index Terms*— Bilateral adaptive evolution, global modeling, pansharpening, transformer.**

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I. Introduction

THE high-spatial-resolution MS (HRMS) images have diverse applications in multiple domains [1], [2], including, but are not limited to, land-cover classification [3], environmental protection [4], and change detection [5], [6]. Nevertheless, due to the physical and technical limitations, existing multispectral (MS) sensors typically observe texture-rich PAN images and low-spatial-resolution MS (LRMS) images separately rather than directly capturing HRMS images. Therefore, the pansharpening technique is devised to produce an HRMS image by fusing the complementary information of an LRMS image and its paired PAN image.

Over the past few decades, there has been a flood of research efforts dedicated to addressing the pansharpening problem, roughly divided into two main categories: traditional methods and deep learning (DL)-based methods [1], [2], [7]. Component substitution (CS)-based approaches [8], [9], multiresolution analysis (MRA)-based methods [10], [11], and variational optimization (VO)-based techniques [12], [13] constitute the primary framework of traditional algorithms. However, these traditional methods suffer from severe spectral or spatial distortions because they are primarily built upon handcrafted prior assumptions, which are inconsistent with actual remote sensing photography.

In recent years, DL-based pansharpening techniques have gained much popularity, showcasing significant performance improvements compared to traditional techniques. PNN [14], inspired by SRCNN [15], is a pioneering effort that introduces a convolutional neural network (CNN) into the pansharpening field. Despite its simplicity, PNN still achieves favorable performance compared to prior conventional algorithms. Subsequently, PanNet [16] achieves a deeper network architecture based on residual learning for pansharpening. Following that, a multitude of CNN-based pansharpening approaches has emerged, such as multiscale architectures [17], [18], [19], adaptive convolution networks [20], [21], and spatial-frequency frameworks [22], [23], [24]. Despite the impressive performance, CNN-based techniques often fail to learn long-range dependencies, owing to the limited receptive fields inherent in convolution operation, hindering further improvements in pansharpening.

Very recently, transformer models, primarily built upon the dot-product self-attention (SA), have gained popularity

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in pansharpening and demonstrate competitive performance compared to CNN-based approaches [25], [26], [27], [28]. Typically, these transformer-based networks are constructed by sequentially stacking multiple transformer blocks, with each block executing SA calculation, as illustrated in Fig. 1(a). Hence, though these methods achieve promising performance gains, the dense SA computations within them still lead to some challenges as follows.

1) The quadratic complexity of the SA brings about huge computational overhead.
2) There exists significant representation redundancy across different layers, as clearly evidenced by the highly similar attention maps shown in Fig. 2.

Such representation redundancy not only complicates network optimization but, more importantly, leads to the insufficient learning of detailed information.

Given these concerns, we propose an efficient bilateral adaptive evolution transformer (BAEFormer) for pansharpening. It is built upon a novel adaptive evolution design, as depicted in Fig. 1(b). Specifically, we first decompose the quadratic complexity in conventional SA into linear-degree height and width computing at the first stage, significantly reducing the computational complexity. Given the innate sparsity of remote sensing images, we devise a novel yet effective neighboring layer-dependent adaptive evolution strategy to update the attention map of two spatial dimensions, thereby avoiding the repetitive SA computation while considering the dynamics toward the evolution of attention weights. Extensive experiments on multiple remote sensing benchmark datasets demonstrate that our proposed BAEFormer outperforms other state-of-the-art methods in both reduced-resolution and real-world full-resolution scenes. The key contributions of this work are summarized as follows.

1) In this work, we first reveal the huge computational overhead and significant representation redundancy resulting from the dense SA computing within existing transformer models. Then, we propose an efficient BAEFormer for pansharpening given these challenges and the inherent sparsity in remote sensing images.
2) We decompose the quadratic complexity in conventional SA into linear-degree height and width computing at the first stage and proceed to apply a novel adaptive evolution mechanism to update the horizontal and vertical attention maps. Such an efficient design dramatically reduces the computational complexity while achieving excellent feature representation capability.
3) Extensive experiments over multiple widely used remote sensing benchmark datasets demonstrate that our BAEFormer outperforms other cutting-edge approaches in both qualitative and quantitative assessments. Moreover, the real-world full-resolution and cross-sensor testing further validate its excellent generalization ability.

II. Related Works

A. Traditional Pansharpening Algorithms

CS, MRA, and VO techniques are three primary families within the realm of traditional pansharpening methods [1], [2], [7]. CS techniques, such as principal component analysis (PCA) [8], partial replacement adaptive CS (PRACS) [30], and hue–intensity–saturation (HIS) [31], exhibit desirable computational efficiency but are prone to generate artifacts. MRA methods integrate multiscale spatial details from PAN images into MS images. The sharpened products resulting from MRA approaches often display apparent spatial distortions. The smoothing filter-based intensity modulation (SFIM) [10], the generalized Laplacian pyramid with modulation transfer function matched filter (MTF-GLP) [32], and the additive wavelet luminance proportional (AWLP) [33] are popular MRA methods. Unlike the aforementioned two categories, VO algorithms approach pansharpening as an optimization problem, wherein the loss function is constructed based on the constraints of predefined priors and then proceeds to attain the desired outcomes through the minimization of the loss function [12], [13], [34]. Methods falling within this category achieve a more favorable balance between preserving spectral and spatial qualities, but their substantial computational demands limit their practical use.

B. Transformer-Based Pansharpening Methods

Transformer models, rooted in the SA mechanism, have garnered significant attention within the computer vision community, thanks to their exceptional capabilities in modeling long-range dependencies. In the realm of pansharpening, Meng et al. [25] pioneer the application of the vision transformer (ViT) to address the long-range dependencies inherent in image patches. This adaptation of transformer architectures showcases the adaptability of these models beyond their original natural language processing domain. To advance pansharpening techniques, INNFormer [28] takes a step further by customizing a transformer architecture, conditioning it on input modalities for feature learning. The subsequent integration of an information-lossless invertible neural module facilitates effective feature fusion, marking a notable stride in the refinement of transformer-based approaches. DR-NET [26] stands out by adopting a regression network architecture founded on the Swin Transformer for pansharpening learning. This approach underscores the versatility of transformer models, demonstrating their efficacy in diverse applications within remote sensing. Furthermore, some hybrid architectures composed of CNN and transformer have been proposed to capture both local and global features [35], [36], [37]. Despite the impressive performance, these methods mainly focus on ameliorating network architectures and neglect the redundancy of the attention map, lacking further exploration of key SA mechanisms within the transformer. Moreover, they often require a high memory footprint and computation budget due to the intricate network designs, contrary to the sparsity of remote sensing images. Very recently, it has been observed that conventional dense SA computations cause representation redundancy, inspiring several efforts to design different global sharing mechanisms to streamline this process [38], [39]. In the realm of pansharpening, Hou et al. [39] advocate for directly evolving the attention weights, thereby eliminating unnecessary and dense SA computations. Although this

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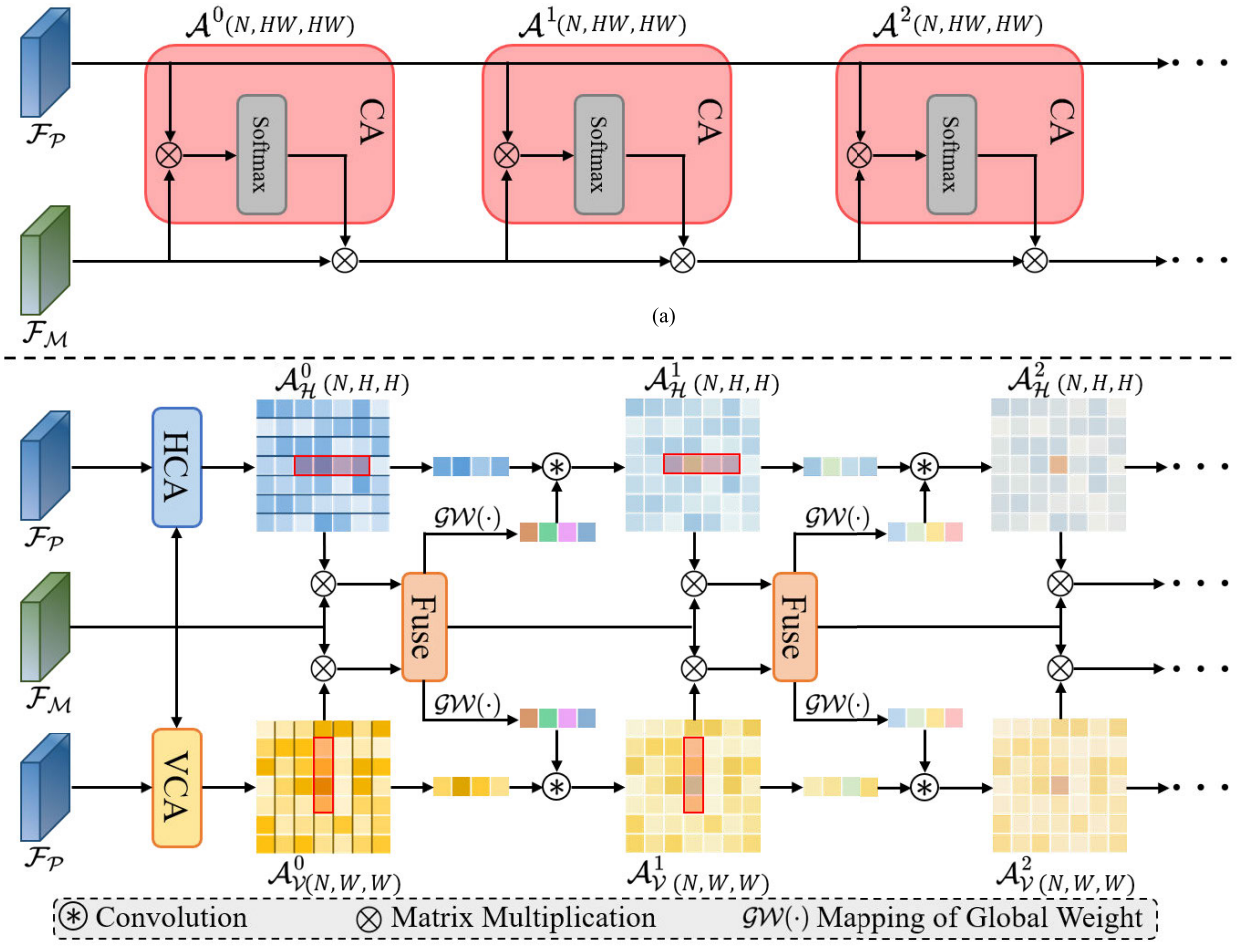


Fig. 1. Comparison between the vanilla cascaded CA and our proposed bilateral adaptive evolution mechanism. Specifically, we separately perform the horizontal and VCA computations (i.e., HCA and VCA; for the detailed structure, refer to Fig. 3) in the first layer, and then, the obtained horizontal attention map $\mathcal{A}^0_\mathcal{H}$ and the vertical attention map $\mathcal{A}^0_\mathcal{V}$ are adaptively evolved based on a shared global weight that is generated through the mapping function $\mathcal{GW}(\cdot)$ (see Fig. 4 for more details). (a) Vanilla cascaded CA architecture. (b) Proposed bilateral adaptive evolution mechanism.

lightweight global modeling design greatly reduces computational overhead, it overlooks the input-adaptive nature of SA weights, potentially leaving room for further improvements. These challenges and opportunities motivate us to explore an efficient task-oriented transformer alternative for pansharpening learning.

III. Methodology

In this section, we first revisit the traditional transformer architecture and then provide an overview of our proposed BAEFormer. Following that, we further delve into the details of two key designs within our model: bilateral attention and adaptive evolution.

A. Revisiting SA

We first revisit the computation of the key SA within the ViT. Given an input token $x \in \mathbb{R}^{1\times C}$ in the input feature $X \in \mathbb{R}^{H\times W\times C}$, it is first projected into three vectors: query ($q \in \mathbb{R}^{1\times d}$), key ($k \in \mathbb{R}^{1\times d}$), and value ($v \in \mathbb{R}^{1\times d}$). Then, by calculating the similarity between the specific query and keys projected by all tokens in X, we can derive a set of attention weights that are applied to weight the values, thereby obtaining the output of the SA module. This process can be formulated as follows:

$$
\begin{aligned}
q_{i,j} &= x_{i,j}W^Q, \quad k_{i,j} = x_{i,j}W^K, \quad v_{i,j} = x_{i,j}W^V \\
\alpha_{s,t}^{i,j} &= \frac{\exp\left(q_{i,j}k_{s,t}\right)}{\sum_{u=0}^{H-1}\sum_{v=0}^{W-1}\exp\left(q_{i,j}k_{u,v}\right)} \\
x'_{i,j} &= \sum_{s=0}^{H-1}\sum_{t=0}^{W-1}\alpha_{s,t}^{i,j}v_{s,t}
\end{aligned} \tag{1}
$$

where $W^Q, W^K, W^V \in R^{C\times d}$ denote the projection matrices and $\alpha_{s,t}^{i,j}$ represent the attention score between $x_{i,j}$ and $x_{s,t}$. $x_{i,j}$, $x'_{i,j}$ is the token located at the ith row and jth column in X and the weighted output corresponding to the $x_{i,j}$ (The subscripts (s, t) and (u, v) also denote the position indices of the corresponding elements.). Pansharpening essentially integrates the complementary information from different modalities of images. Thus, we employ the cross-attention (CA) computation to perform the cross-modal interactions based on the SA.

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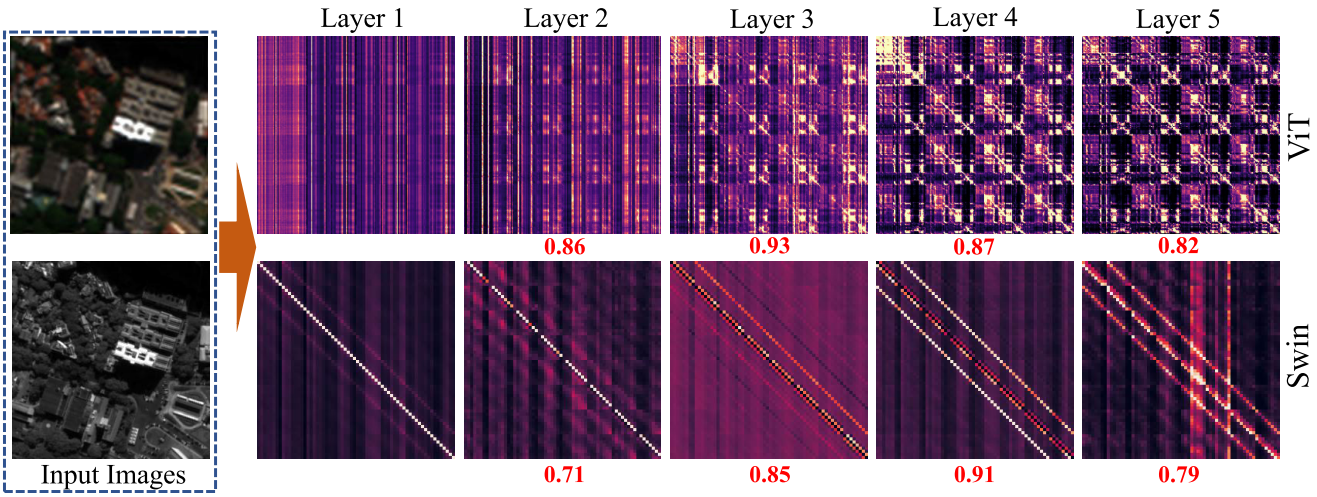


Fig. 2. Analysis of the attention map correlation. It visualizes the averaged attention maps from the ViT [25] and Swin Transformer (Swin) [29] across multiple heads. The red numbers beneath attention maps indicate the cosine similarity between the attention maps of the ith and (i-1)th layers. The visualization reveals that both ViT and Swin exhibit highly similar attention maps across neighboring layers, especially in intermediate layers.

B. Overall Architecture

As detailed in Fig. 3, we first upsample the original observed MS image $\mathcal{M} \in \mathbb{R}^{h\times w\times c}(h = (H/4)$ and $w = (W/4))$ to the same spatial resolution as PAN image. Then, the upsampled MS image $\mathcal{M}^U \in \mathbb{R}^{H\times W\times c}$ and the corresponding PAN image $\mathcal{P} \in \mathbb{R}^{H\times W}$ are projected into feature space using several convolutional layers as follows:

$$\mathcal{F}_{\mathcal{P}} = \text{Proj}(\mathcal{P}), \mathcal{F}_{\mathcal{M}} = \text{Proj}(\mathcal{M}^U) \tag{2}$$

where Proj$(\cdot)$ represents the mapping function of the convolutional layers.

Subsequently, we decompose the CA calculation into height and width computing in the first block. Specifically, we perform the horizontal and vertical multihead CA (MCA) calculations, respectively, followed by a bilateral feature information aggregation (BFIA) module to integrate the outcomes from these two attention modules to obtain the global feature representation. The detailed process is written as follows:

$$\begin{aligned}
\mathcal{Z}^0_{\mathcal{H}} &= \mathcal{A}^0_{\mathcal{H}} \otimes \mathcal{F}_{\mathcal{M}} = \text{HCA}(\mathcal{F}_{\mathcal{P}}, \mathcal{F}_{\mathcal{M}}) \otimes \mathcal{F}_{\mathcal{M}} \\
\mathcal{Z}^0_{\mathcal{V}} &= \mathcal{A}^0_{\mathcal{V}} \otimes \mathcal{F}_{\mathcal{M}} = \text{VCA}(\mathcal{F}_{\mathcal{P}}, \mathcal{F}_{\mathcal{M}}) \otimes \mathcal{F}_{\mathcal{M}} \\
\mathcal{Z}^0 &= \varphi\left(\mathcal{Z}^0_{\mathcal{H}}, \mathcal{Z}^0_{\mathcal{V}}\right)
\end{aligned} \tag{3}$$

where $\otimes$ denotes the matrix multiplication. HCA$(\cdot)$ and VCA$(\cdot)$ represent the horizontal and vertical MCA calculations, $\mathcal{A}^0_{\mathcal{H}}$ and $\mathcal{A}^0_{\mathcal{V}}$ are the corresponding MCA maps. $\mathcal{Z}^0_{\mathcal{H}}$ and $\mathcal{Z}^0_{\mathcal{V}}$ denote the outcomes corresponding to the horizontal and vertical MCA modules. $\varphi(\cdot)$ denotes the mapping function of the BFIA block. $\mathcal{Z}^0$ is the final output of the first block.

Following that, multiple BAEFormer blocks are employed to perform the feature learning for generating the desired HRMS images, in which the key bilateral adaptive evolution mechanism is applied to update the attention map. Mathematically, it can be formulated as

$$\begin{aligned}
\mathcal{Z}^1_{\mathcal{H}} &= \mathcal{A}^1_{\mathcal{H}} \otimes \mathcal{Z}^0 = \phi(\mathcal{Z}^0, \mathcal{A}^0_{\mathcal{H}}) \otimes \mathcal{Z}^0 \\
\mathcal{Z}^1_{\mathcal{V}} &= \mathcal{A}^1_{\mathcal{V}} \otimes \mathcal{Z}^0 = \phi(\mathcal{Z}^0, \mathcal{A}^0_{\mathcal{V}}) \otimes \mathcal{Z}^0 \\
\mathcal{Z}^1 &= \varphi\left(\mathcal{Z}^1_{\mathcal{H}}, \mathcal{Z}^1_{\mathcal{V}}\right) \\
&\cdots \\
\mathcal{Z}^m_{\mathcal{H}} &= \mathcal{A}^m_{\mathcal{H}} \otimes \mathcal{Z}^{m-1} = \phi\left(\mathcal{Z}^{m-1}, \mathcal{A}^{m-1}_{\mathcal{H}}\right) \otimes \mathcal{Z}^{m-1} \\
\mathcal{Z}^m_{\mathcal{V}} &= \mathcal{A}^m_{\mathcal{V}} \otimes \mathcal{Z}^{m-1} = \phi\left(\mathcal{Z}^{m-1}, \mathcal{A}^{m-1}_{\mathcal{V}}\right) \otimes \mathcal{Z}^{m-1} \\
\mathcal{Z}^m &= \varphi\left(\mathcal{Z}^{\text{m}}_{\mathcal{H}}, \mathcal{Z}^{\text{m}}_{\mathcal{V}}\right)
\end{aligned} \tag{4}$$

where $\phi(\cdot)$ denotes the mapping function of the proposed horizontal adaptive evolution (HAE) or vertical adaptive evolution (VAE). m is the number of the BAEFormer blocks. $\mathcal{A}^{\text{k}}_{\mathcal{H}}$ and $\mathcal{A}^k_{\mathcal{V}}, k \in \{1, 2, \ldots, m\}$, represent the horizontal and vertical evolution attention maps.

Finally, by performing the convolution layers on the output of the mth BAEFormer block and adding the upsampled MS image, we can obtain the pansharpened product of our model

$$O = \text{Conv}(\mathcal{Z}^m) + \mathcal{M}^U \tag{5}$$

where Conv$(\cdot)$ denotes the convolution layers.

C. Proposed Core Mechanisms

1) Bilateral Attention: As displayed in Fig. 3, the proposed bilateral attention mechanism performs the horizontal and vertical MCA in parallel to capture the elementwise long-range dependencies. Given an input token $x_{i,j}$ located at the ith row and jth column in the input feature X, the horizontal MCA computes attention scores between itself and other tokens within the same row in feature Y projected by X, rather than all tokens. Then, the attention output for $x_{i,j}$ can be computed by applying the obtained weights to the corresponding tokens in the nth head, as outlined in the following:

$$\begin{aligned}
q^n_{i,j} &= x_{i,j} W^Q_n, \quad k^n_{i,j} = y_{i,j} W^K_n, \quad v^n_{i,j} = y_{i,j} W^V_n \\
\alpha^{i,j,n}_{i,t} &= \frac{\exp\left(q^n_{i,j} k^n_{i,t}\right)}{\sum_{v=0}^{W-1} \exp\left(q^n_{i,j} k^n_{i,v}\right)} \\
x^n_{i,j} &= \sum_{t=0}^{W-1} \alpha^{i,j,n}_{i,t} v^n_{i,t}.
\end{aligned} \tag{6}$$

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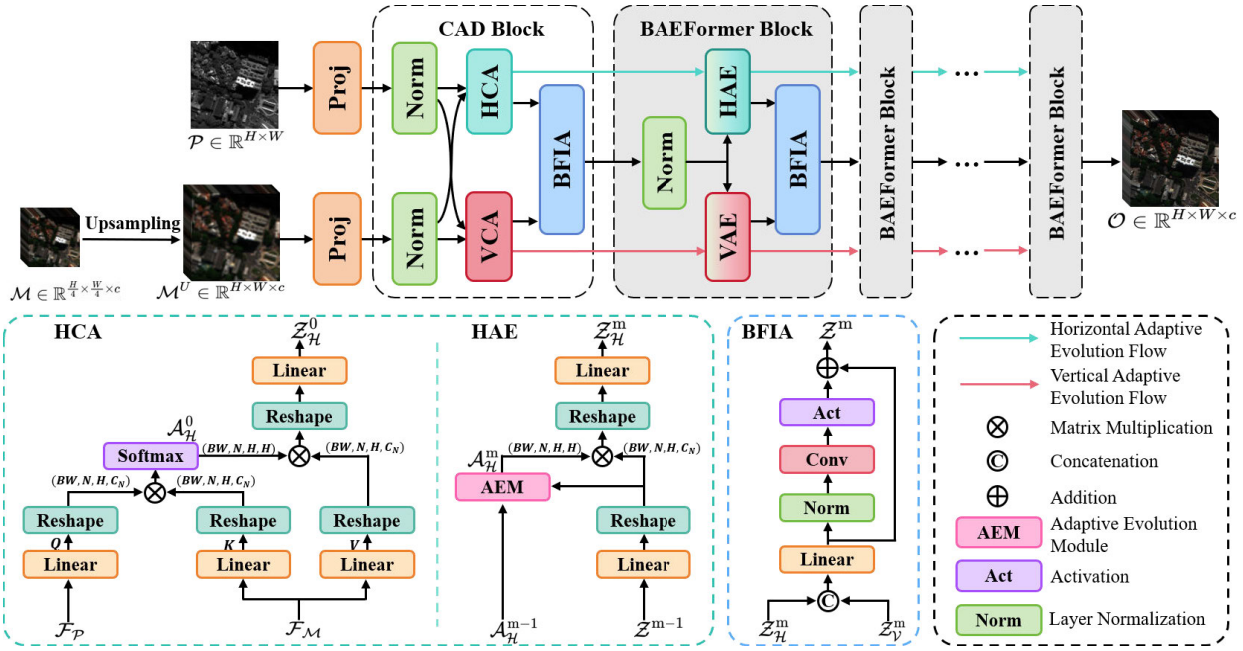


Fig. 3. Detailed network architecture of the proposed BAEFormer, consisting of a CA decomposing (CAD) block and several proposed BAEFormer blocks. Each block contains a horizontal and a vertical branch. The VCA and VAE structures are symmetrically implemented similar to the horizontal designs, so we have omitted them.

By concatenating the results from N heads along the channel dimension, we can obtain the horizontal MCA output for the token $x_{i,j}$

$$x_{i,j}^{h} = \mathrm{Cat}\left(x_{i,j}^{1}, x_{i,j}^{2}, \ldots, x_{i,j}^{N}\right). \tag{7}$$

Likewise, the vertical MCA calculation can be defined as follows:

$$\begin{aligned} \alpha_{s,j}^{i,j,n} &= \frac{\exp\left(q_{i,j}^{n} k_{s,j}^{n}\right)}{\sum_{u=0}^{H-1} \exp\left(q_{i,j}^{n} k_{u,j}^{n}\right)} \\ x_{i,j}^{n} &= \sum_{s=0}^{H-1} \alpha_{s,j}^{i,j,n} v_{s,j}^{n} \\ x_{i,j}^{v} &= \mathrm{Cat}\left(x_{i,j}^{1}, x_{i,j}^{2}, \ldots, x_{i,j}^{N}\right). \end{aligned} \tag{8}$$

Afterward, by fusing the horizontal and vertical results via BFIA, we can obtain the final attention output for token $x_{i,j}$

$$x_{i,j}' = \varphi\left(x_{i,j}^{h}, x_{i,j}^{v}\right). \tag{9}$$

2) Adaptive Evolution: As displayed in Fig. 3, the HAE (or VAE) within the BAEFormer blocks utilizes a novel adaptive evolution module (AEM) to evolve the attention maps, thereby replacing the dense CA computations in conventional transformer designs. Fig. 4 presents the detailed flowchart of the designed AEM. Given a group of multihead attention maps $A \in \mathbb{R}^{N\times M\times M}$, where N denotes the number of heads, and M refers to the size of the attention map, in which $M = H$ and $M = W$ for the horizontal and vertical cases, respectively, the designed AEM generates a series of adaptive kernels for each multihead attention patch conditioned on the neighboring block's output.

Specifically, we first perform a $1 \times k$ convolution layer coupled with nonlinear activation on an extracted attention patch. Followed by, we further apply a depthwise convolution layer to learn the feature information. This process can be formulated as follows:

$$K_{A}^{i,j} = \mathrm{DW}(\psi(\mathrm{Conv}_{1\times k}(A_{1\times k}^{i,j}))) \tag{10}$$

where $A_{1\times k}^{i,j}$ denotes the attention patch located at the ith row and jth column in the attention map with size $1 \times k$. $\psi(\cdot)$ and $\mathrm{Conv}_{1\times k}(\cdot)$ are the nonlinear activation function and the $1 \times k$ convolution operator. $K_{A}^{i,j}$ represents the resulting advanced attention patch.

Then, we apply a global average pooling operator to aggregate the global information of the neighboring BAEFormer block's output and then use linear layers to learn the weight information as follows:

$$K_{Z} = W(\mathrm{GAP}(Z^{m-1})) \tag{11}$$

where $\mathrm{GAP}(\cdot)$ and $W(\cdot)$ denote the mapping of the global average pooling operator and the linear layer, respectively. K_Z represents the aggregated weight information.

After that, by performing the elementwise multiplication between $K_{A}^{i,j}$ and K_Z, we can obtain the final adaptive evolution kernel specific to the attention patch $A_{1\times k}^{i,j}$

$$K^{i,j} = K_{A}^{i,j} \odot K_{Z} \tag{12}$$

where $\odot$ denotes the elementwise multiplication. $K^{i,j}$ is the final adaptive evolution kernel.

Finally, the attention weights of the mth layer can be updated by applying the obtained adaptive evolution kernel to the attention map of the m-1th layer as follows:

$$\alpha_{i,j}^{m} = \sum_{k=0}^{K-1} \alpha_{i,j-\lfloor \frac{K}{2} \rfloor + k}^{m-1} K_{k}^{i,j} \tag{13}$$

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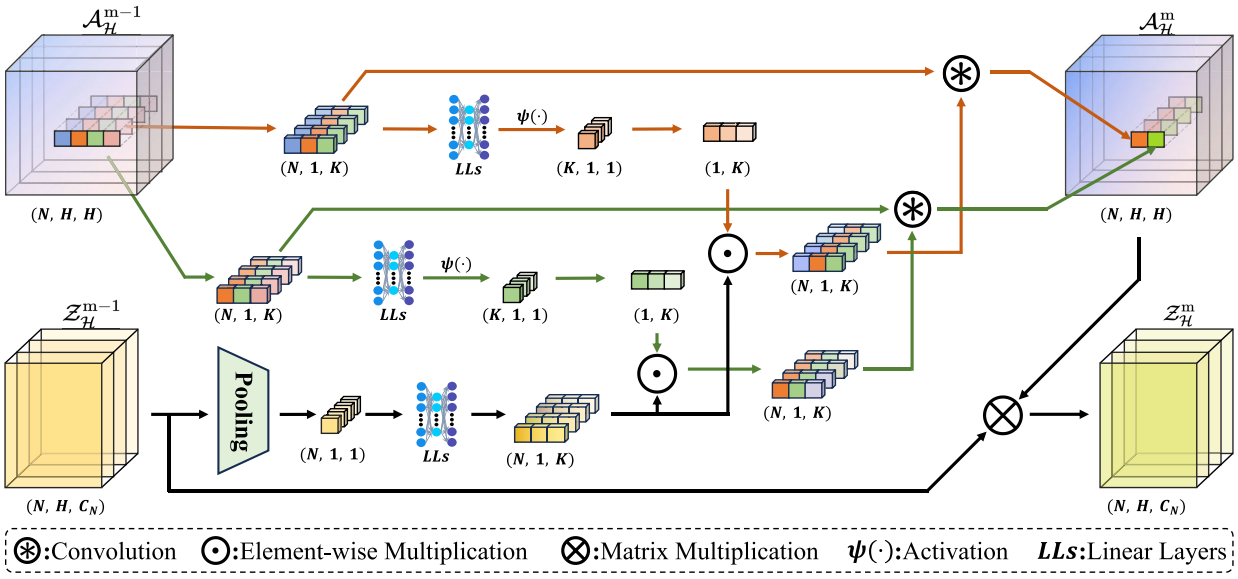


Fig. 4. Detailed flowchart of our proposed AEM. Notably, we take the horizontal attention evolution as an instance for illustration, while we have omitted the vertical attention evolution since it is symmetrically performed similar to the horizontal one.

where $\lfloor\cdot\rfloor$ denotes rounding down operation. $\alpha_{i,j}^{m}$ is the evolved attention value within the mth layer.

D. Network Optimization

We employ two loss terms in our approach, including the reconstruction loss $\mathcal{L}_{\text{rec}}$ and the structure loss $\mathcal{L}_{\text{str}}$, to optimize the network parameters during the training process. The complete loss function can be expressed as

$$\mathbf{L}_{\text{total}} = \mathbf{L}_{\text{rec}} + \alpha \mathbf{L}_{\text{str}} \tag{14}$$

where the hyperparameter α is introduced to strike a balance between the overall performance and the preservation of texture details. In particular, we utilize the straightforward L_1 loss to calculate the reconstruction loss, defined as follows:

$$\mathbf{L}_{\text{rec}} = \|\mathbf{H}_s - \mathbf{GT}\|_1 \tag{15}$$

where $\mathbf{H}_s$ and $\mathbf{GT}$ represent the pansharpened result and ground truth, respectively. $||\cdot||_1$ denotes the L_1 norm.

In addition to the reconstruction loss, we incorporate the structural similarity (SSIM) as an additional loss function, defined as follows:

$$\mathbf{L}_{\text{str}} = \|1 - \text{SSIM}(\mathbf{H}_s, \mathbf{GT})\|_1 \tag{16}$$

where $\text{SSIM}(\cdot)$ represents the SSIM function.

IV. Experiments

In this section, we first detail the experimental settings, including the dataset simulation, metrics, training details, and baseline. Then, we evaluate the effectiveness of our model on both reduced-resolution and real-world full-resolution scenes and further investigate its generalization capability through cross-sensor testing. Moreover, we perform an ablation study to illuminate the efficacy of each key design within the proposed BAEFormer and its superiority over the traditional transformer architecture.

A. Datasets and Experimental Settings

1) Dataset Simulation: We evaluate the effectiveness of our pansharpening model across three well-recognized commercial satellites: WorldView3 (WV3), GaoFen2 (GF2), and WorldView2 (WV2). Each dataset comprises thousands of image pairs dedicated to network training, validation, and testing. For the training set, the spatial resolution of LRMS, PAN, and GT is 64 × 64, while it is 16 × 16 for MS. In terms of the testing set, the reduced-resolution testing dataset has a resolution of 256 × 256 for LRMS, PAN, and GT, and 64 × 64 for MS, while the full-resolution testing samples adopt 512 × 512 for LRMS and PAN, and 128 × 128 for MS. For more details about data simulation, refer to [7], and all the training and testing data used in this work are accessible on the public website.[1]

2) Metrics: In the experiments, we employ several widely used image quality assessment (IQA) metrics to assess the model performance. The spectral angle mapper (SAM) [40], the dimensionless global error in synthesis (ERGAS) [41], the spatial correlation coefficient (SCC) [42], and the Q2n (Q8 for eight-band datasets and Q4 for four-band datasets) [43] are adopted for the reduced-resolution performance assessment. For full-resolution evaluation, moreover, three nonreference metrics, namely, the hybrid quality with no reference (HQNR) index, the spectral distortion D_λ index, and the spatial distortion D_s index [44], are selected. In all quantitative comparisons, the up ↑ indicates that higher metric values represent better performance, while the down arrow ↓ denotes that lower metric values correspond to better performance.

3) Training Details: All DL models are implemented through PyTorch 2.0 and Python 3.10 using a Linux operating system with an NVIDIA RTX4090 GPU. We utilize the Adam algorithm for network optimization, setting beta values to (0.9, 0.999) and employing a weight decay of 0.1. The minibatch size and the initial learning rate are set to 16 and 3×10^{-4},

[1] https://github.com/liangjiandeng/PanCollection

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TABLE I

QUANTITATIVE COMPARISON WITH THE STATE-OF-THE-ART METHODS ON REDUCED RESOLUTION (RED: BEST; BLUE: SECOND BEST)

Method	WV3				GF2			
	SAM↓	ERGAS↓	Q8↑	SCC↑	SAM↓	ERGAS↓	Q4↑	SCC↑
BT-H [48]	4.9198	4.5789	0.8324	0.9252	1.6488	1.5280	0.9177	0.9570
BDSD-PC [49]	5.4293	4.6976	0.8294	0.9081	1.6813	1.6667	0.8922	0.9521
MTF-GLP-HPM-R [11]	5.3383	5.2301	0.8346	0.8892	1.6502	1.5875	0.8996	0.9480
LRTCFPan [13]	4.7367	4.3153	0.8463	0.9273	1.2977	1.2723	0.9350	0.9637
PNN [14]	3.6798	2.6819	0.8929	0.9761	1.0477	1.0572	0.9604	0.9772
FusionNet [45]	3.3252	2.4666	0.9044	0.9807	0.9735	0.9878	0.9641	0.9806
GPPNN [46]	3.0554	2.3056	0.9143	0.9837	0.7972	0.7107	0.9791	0.9897
LAGNet [20]	3.1042	2.2999	0.9098	0.9838	0.7859	0.6869	0.9804	0.9906
ViTPan [25]	3.0923	2.3329	0.9053	0.9848	0.8027	0.7015	0.9771	0.9907
INNFormer [28]	3.2174	2.3604	0.9117	0.9831	0.7875	0.6619	0.9801	0.9919
HyperTransformer [27]	3.4563	2.5995	0.9013	0.9783	0.8539	0.7881	0.9759	0.9879
GuidedNet [18]	2.9838	2.2711	0.9177	0.9844	0.8268	0.7191	0.9798	0.9901
Fourmer [47]	3.2363	2.4189	0.9108	0.9840	0.9757	0.8845	0.9698	0.9870
Ours	2.8974	2.1336	0.9205	0.9868	0.7339	0.6602	0.9816	0.9909

TABLE II

COMPARISON OF NETWORK PARAMETERS AND FLOPS OF DIFFERENT DL-BASED APPROACHES ON THE WV3 DATASET. NOTE THAT HYPER DENOTES THE HYPERTRANSFORMER

Metrics	PNN	FusionNet	GPPNN	LAGNet	ViTPan	INNFormer	Hyper	GuidedNet	Fourmer	**Ours**
Params (M)	0.10	0.05	0.24	0.15	15.91	0.14	34.36	1.13	5.21	0.34
FLOPs (G)	0.30	0.32	0.98	0.53	4.40	3.89	13.67	2.29	5.12	1.69

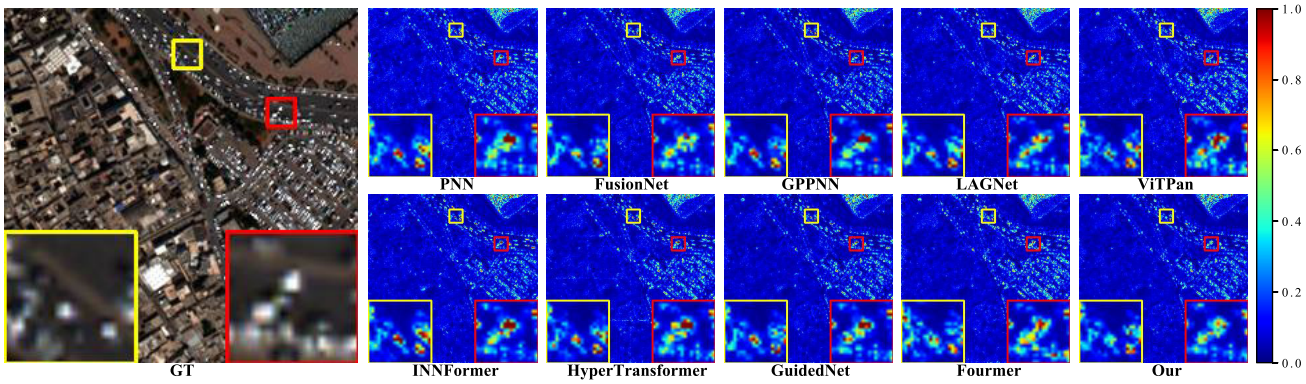


Fig. 5. Comparison of the error maps between our model and other cutting-edge methods over the WV3 dataset.

respectively. The learning rate decay is applied by multiplying 0.1 at 500 and 1000 epochs, with training concluding after 1500 epochs. In all experiments, the hyperparameter λ in the loss function remains constant at 0.1, and we adopt four BAEFormer modules in our model.

4) Baselines: We choose nine cutting-edge DL techniques, including PNN [14], FusionNet [45], GPPNN [46], LAGNet [20], ViTPan [25], INNFormer [28], HyperTransformer [27], GuidedNet [18], and Fourmer [47], to showcase the superiority of our model. In addition, four representative traditional algorithms are selected for a wide comparison: BT-H [48], BDSD-PC [49], MTF-GLP-HPM-P [11], and LRTCFPan [13].

B. Comparison With State-of-the-Art Methods

1) Evaluation on Reduced-Resolution Scene: We initially conducted a reduced-resolution assessment to gauge the disparity between HRMS images predicted by various methods and the GT images. The quantitative outcomes across WV3 and GF2 datasets are tabulated in Table I, where superior performance is highlighted in red and the second in blue. The results in Table I unmistakably showcase our model's superiority across all evaluation metrics compared to other methodologies, affirming its prowess. Notably, our model consistently outperforms other attention-based techniques (e.g., ViTPan, INNFprmer, and HyperTransformer) in terms of both spatial detail reconstruction and spectral information preservation, as evidenced by the best quantitative scores across multiple datasets. In addition, our model also presents the favorable memory footprint and computational cost against other transformer-based counterparts, as reported in Table II, thereby yielding the desired performance–efficiency balance. The commendable performance of our BAEFormer model is attributed to its adeptness in handling the sparsity inherent

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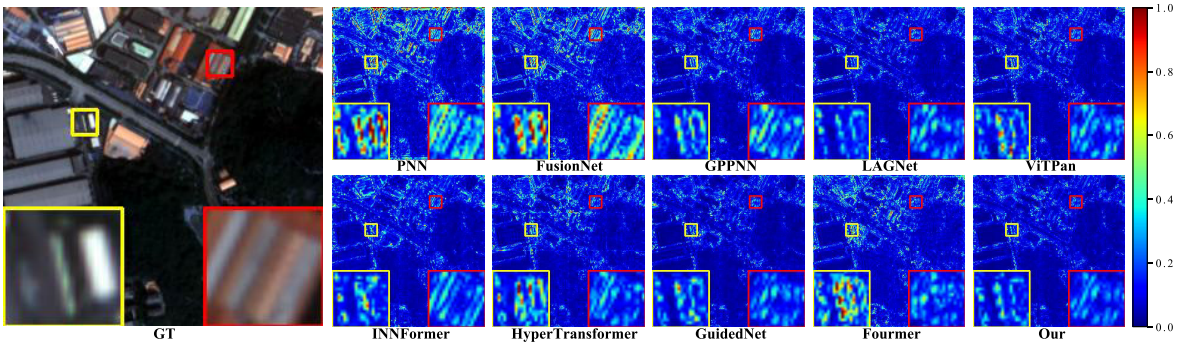


Fig. 6. Comparison of the error maps between our model and other cutting-edge methods over the GF2 dataset.

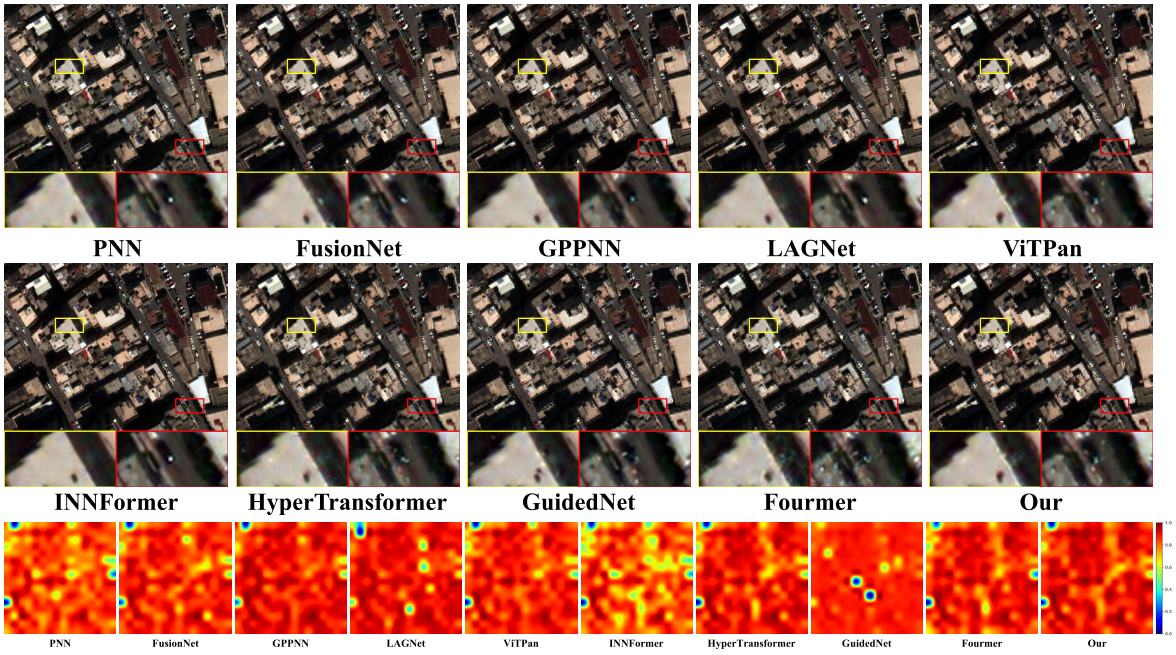


Fig. 7. Visual comparison with state-of-the-art methods on the full-resolution WV3 dataset.

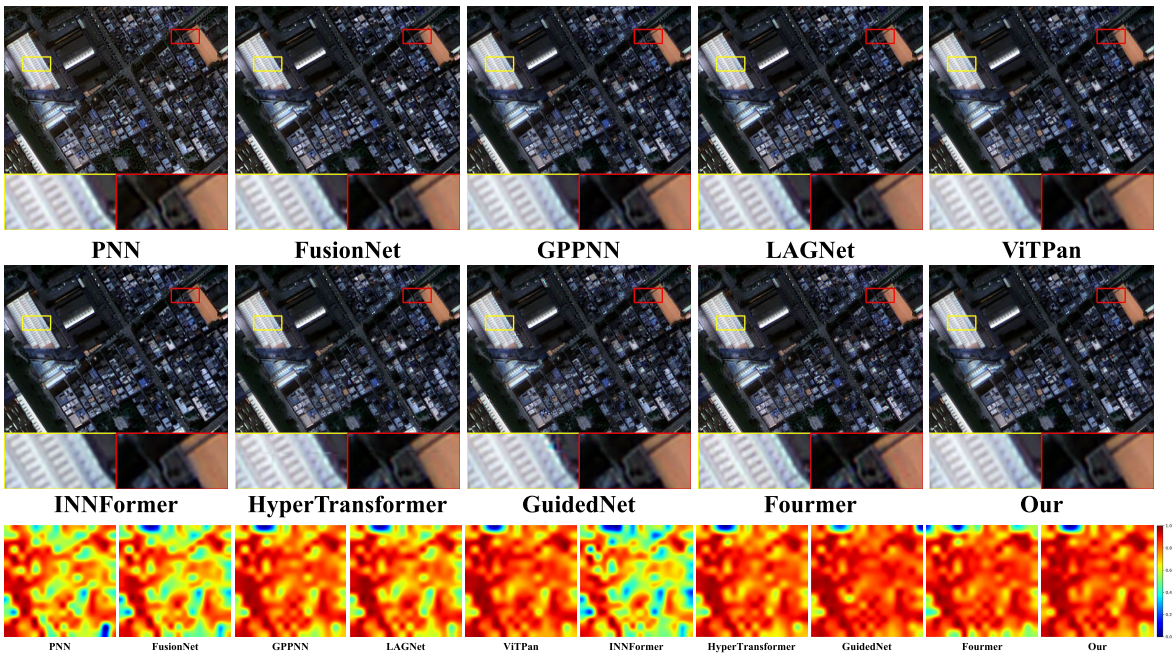


Fig. 8. Visual comparison with state-of-the-art methods on the full-resolution GF2 dataset.

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TABLE III
QUANTITATIVE COMPARISON WITH THE STATE-OF-THE-ART METHODS ON FULL RESOLUTION (RED: BEST; BLUE: SECOND BEST)

Method	WV3 $D_\lambda\downarrow$	$D_s\downarrow$	HQNR↑
BT-H [48]	0.0574	0.0810	0.8670
BDSD-PC [49]	0.0625	0.0730	0.8698
MTF-GLP-HPM-R [11]	0.0206	0.0630	0.9180
LRTCFPan [13]	0.0176	0.0528	0.9307
PNN [14]	0.0213	0.0428	0.9369
FusionNet [45]	0.0239	0.0364	0.9406
GPPNN [46]	0.0281	0.0378	0.9352
LAGNet [20]	0.0368	0.0418	0.9230
ViTPan [25]	0.0179	0.0341	0.9486
INNFormer [28]	0.0550	0.0679	0.8815
HyperTransformer [27]	0.0274	0.0386	0.9462
GuidedNet [18]	0.0291	0.0410	0.9312
Fourmer [47]	0.0224	0.0347	0.9437
Ours	0.0158	0.0332	0.9515
Method	**GF2** $D_\lambda\downarrow$	$D_s\downarrow$	HQNR↑
BT-H [48]	0.0602	0.1313	0.8165
BDSD-PC [49]	0.0759	0.1548	0.7812
MTF-GLP-HPM-R [11]	0.0336	0.1404	0.8309
LRTCFPan [13]	0.0325	0.0896	0.8806
PNN [14]	0.0317	0.0943	0.8771
FusionNet [45]	0.0350	0.1013	0.8673
GPPNN [46]	0.0229	0.0665	0.9122
LAGNet [20]	0.0284	0.0792	0.8947
ViTPan [25]	0.0304	0.0507	0.9203
INNFormer [28]	0.0609	0.1096	0.8360
HyperTransformer [27]	0.0231	0.0552	0.9230
GuidedNet [18]	0.0209	0.0573	0.9229
Fourmer [47]	0.0470	0.0380	0.9166
Ours	0.0224	0.0495	0.9292

in remote sensing images, thereby mitigating redundancy in dense attention computation. Crucially, the devised attention evolution mechanism aptly accommodates the evolving trends within attention maps across different layers. This adaptability enables our model to adeptly capture crucial information emerging during the pansharpening process with greater precision and fluidity. Furthermore, the comparison of error analysis against GT images, as presented in Figs. 5 and 6, vividly showcases the discernible advantages of our proposed pansharpening framework over other DL-based methods. It is evident from Figs. 5 and 6 that our model yields results that closely resemble the GT images as evidenced by its dark blue error maps, exhibiting superior spatial texture reconstruction.

2) Evaluation on Full-Resolution Scene: We conduct further experiments on high-resolution (HR) examples to assess our model's real-world performance and adaptability. Specifically, we apply the pretrained model, derived from reduced-resolution data, directly to various full-resolution samples. It is essential to note that due to the unavailability of GT images, we rely on three widely recognized no-reference evaluation metrics to assess the full-resolution performance. The quantitative results for all compared methods across the WV3 and GF2 datasets are consolidated into Table III. As depicted in Table III, our proposed framework demonstrates

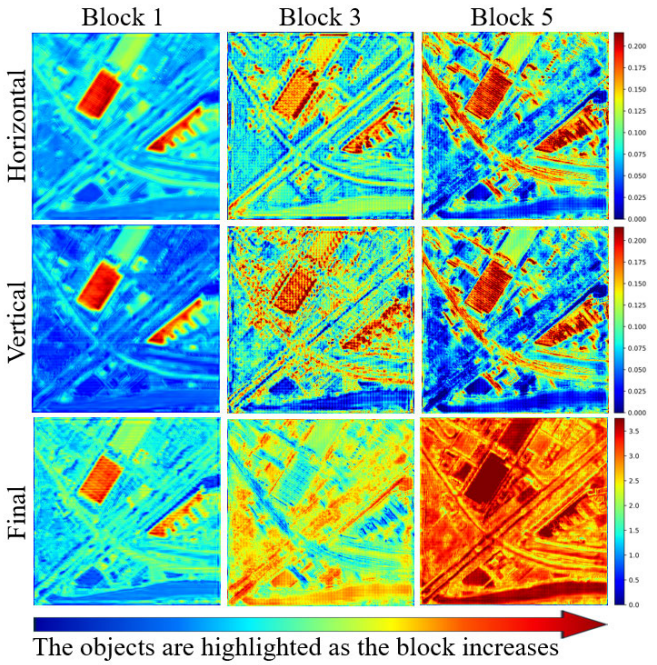


Fig. 9. Feature maps of the horizontal, VAE branch, and final outputs in the first, third, and fifth BAEFormer blocks. The highlighted objects showcase the efficacy of our design as the block increases.

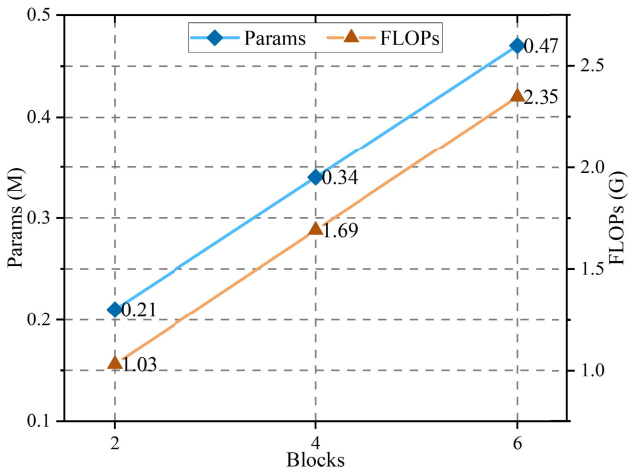


Fig. 10. Relationship between computational overhead and block number. It is clear that both the parameters and FLOPs increase linearly with the number of blocks.

nearly optimal results across all indices, underscoring its exceptional generalization capabilities in contrast to traditional and state-of-the-art DL-based pansharpening technologies. The visual comparisons with state-of-the-art methods on the full-resolution WV3 and GF2 datasets are presented in Figs. 7 and 8, in which our method exhibits clearer texture details. In addition, the second raw of Figs. 7 and 8 further visualizes the corresponding HQNR maps of all compared DL-based techniques. It can be observed that our method achieves the optimal outcomes, as indicated by the dark-colored HQNR maps with sparse bright spots, underscoring high alignment with the quantitative results in Table III. Both the quantitative assessments and qualitative analyses collectively validate the practical applicability of our model in real-world scenarios.

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TABLE IV
QUANTITATIVE COMPARISON WITH THE STATE-OF-THE-ART METHODS ON THE WV2 DATASET. (RED: BEST; BLUE: SECOND BEST). NOTE THAT HYPER DENOTES THE HYPERTRANSFORMER

Metrics	PNN	FusionNet	GPPNN	LAGNet	ViTPan	INNFormer	Hyper	GuidedNet	Fourmer	**Ours**
SAM↓	7.1158	6.4257	6.8295	6.9545	6.7484	6.2793	6.0472	5.6789	6.2241	5.6342
ERGAS↓	5.6152	5.1363	5.1743	5.3262	4.8156	4.6189	4.7806	4.5329	4.9145	4.5241
Q8↑	0.7619	0.7961	0.7942	0.8054	0.8233	0.8237	0.8127	0.8261	0.8214	0.8286
SCC↑	0.8782	0.8746	0.9086	0.9125	0.9102	0.9170	0.9018	0.9049	0.8980	0.9136

TABLE V
EFFECT OF DIFFERENT ATTENTION MECHANISMS AND THE BLOCK NUMBERS. (a) EFFECT OF ATTENTION MECHANISMS. (b) EFFECT OF THE BLOCK NUMBERS

(a) Effect of attention mechanisms.

Attention	SAM↓	ERGAS↓	Q8↑	SCC↑
AEA	2.8974	2.1336	0.9205	0.9868
SCA	5.2793	4.9011	0.7437	0.9247
CA	4.7461	3.5742	0.8675	0.9615

(b) Effect of the block numbers.

Blocks	SAM↓	ERGAS↓	Q8↑	SCC↑
2	3.0115	2.2090	0.9173	0.9851
4	2.8974	2.1336	0.9205	0.9868
6	2.8973	2.1271	0.9212	0.9867

C. Visualization of Feature Maps in Different Block

To investigate the influence of the proposed bilateral adaptive evolution mechanism, we present visualizations of feature maps within various blocks, as illustrated in Fig. 9. Specifically, we take an example from the GF2 dataset, where the horizontal, vertical, and final outputs represent the respective outcomes from the bilateral adaptive evolution units and the fused output, respectively. The features in the aforementioned two lines illustrate that the attention evolution block for rows and columns adeptly captures key information in each layer. Simultaneously, it effectively models the gradual evolution of attention across different layers, providing evidence of their contribution to exploring diversity and specificity. The integration of these diverse features results in an informative final output feature map.

D. Cross-Sensor Generalization Capability

We further validate the generalization capability of our BAEFormer by performing the cross-sensor evaluation. Specifically, we assess the performance of all DL-based models that are trained on the WV3 dataset without any fine-tuning on WV2 testing examples. This approach is reasonable because the WV2 and WV3 datasets share the same number of spectral bands, while there are slight disparities in their spatial resolution. Table IV presents the quantitative outcomes of all comparative methods. Manifestly, our proposed model consistently attains results approaching superiority across all metrics, demonstrating its commendable applicability and resilience.

E. Ablation Study

We perform the ablation experiments on the WV3 dataset to analyze the impact of various ingredients, including the proposed core design and hyperparameters.

1) Effect of the Different Attention Mechanisms: To explore the effectiveness of our proposed bilateral adaptive evolution mechanism, we compare the differences of three attention variants. We take the model with five transformer blocks for comparison. The specific model variants are given as follows.

1) *Baseline:* Conduct the CA computation in the initial block, and then, apply the devised AEM to update the attention maps in the remaining blocks, denoted as adaptive evolution attention (AEA).
2) *Variant-1:* Remove the AEM, and directly share the CA map from the initial block with the remaining blocks without updating, denoted as shared CA (SCA).
3) *Variant-2:* Perform the CA computations in each block, denoted as CA.

Table V(a) displays the corresponding quantitative comparisons. It is evident that the model incorporating our proposed AEA achieves competitive performance, showcasing its efficacy compared to the other two variants.

2) Effect of the Block Numbers: To investigate the impact of the number of proposed BAEFormer on model performance, we adopt different numbers of BAEFormer blocks for comparison and show the results in Table V(b). It is clear that even with only two BAEFormer blocks, the performance of our model still surpasses most DL methods. With the stacking of the BAEFormer block, the performance of our framework would be superior. As illustrated in Fig. 10, the network parameters and FLOPs increase linearly with the number of blocks. To achieve a better performance–efficiency tradeoff, we use four BAEFormer blocks as the default setting in our main experiments.

V. CONCLUSION

In this work, our proposed model, the BAEFormer, introduces two key mechanisms: bilateral attention computation and adaptive attention evolution. We successfully decompose

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the traditional quadratic complexity SA into linear-degree height and width computations at the initial stage, substantially reducing computational complexity. In addition, by considering data-specific properties, we present an innovative neighboring layer-dependent strategy for the adaptive updating of the attention map in two spatial dimensions. This approach mitigates repetitive SA computation and accounts for the dynamic evolution of attention weights. Notably, our model, BAEFormer, demonstrates superior performance over various state-of-the-art pansharpening methods when applied to diverse remote sensing datasets. Furthermore, it achieves these results with a reduced number of network parameters and computational requirements.

REFERENCES

[1] X. Meng, H. Shen, H. Li, L. Zhang, and R. Fu, "Review of the pansharpening methods for remote sensing images based on the idea of meta-analysis: Practical discussion and challenges," *Inf. Fusion*, vol. 46, pp. 102–113, Mar. 2019.

[2] G. Vivone et al., "A new benchmark based on recent advances in multispectral pansharpening: Revisiting pansharpening with classical and emerging pansharpening methods," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 1, pp. 53–81, Mar. 2021.

[3] I. R. Farah, W. Boulila, K. S. Ettabaa, and M. B. Ahmed, "Multiapproach system based on fusion of multispectral images for land-cover classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 12, pp. 4153–4161, Dec. 2008.

[4] T. Berhane et al., "Decision-tree, rule-based, and random forest classification of high-resolution multispectral imagery for wetland mapping and inventory," *Remote Sens.*, vol. 10, no. 4, p. 580, Apr. 2018.

[5] H. Zhang, M. Gong, P. Zhang, L. Su, and J. Shi, "Feature-level change detection using deep representation and feature change analysis for multispectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 11, pp. 1666–1670, Nov. 2016.

[6] L. Mou, L. Bruzzone, and X. X. Zhu, "Learning spectral–spatial–temporal features via a recurrent convolutional neural network for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 924–935, Feb. 2019.

[7] L.-J. Deng et al., "Machine learning in pansharpening: A benchmark, from shallow to deep networks," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 3, pp. 279–315, Sep. 2022.

[8] P. Kwarteng and A. Chavez, "Extracting spectral contrast in Landsat thematic mapper image data using selective principal component analysis," *Photogramm. Eng. Remote Sens.*, vol. 55, no. 1, pp. 339–348, 1989.

[9] A. Garzelli, F. Nencini, and L. Capobianco, "Optimal MMSE pan sharpening of very high resolution multispectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 1, pp. 228–236, Jan. 2008.

[10] G. J. Liu, "Smoothing filter-based intensity modulation: A spectral preserve image fusion technique for improving spatial details," *Int. J. Remote Sens.*, vol. 21, no. 18, pp. 3461–3472, 2000.

[11] G. Vivone, R. Restaino, and J. Chanussot, "A regression-based high-pass modulation pansharpening approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 984–996, Feb. 2018.

[12] J.-L. Xiao, T.-Z. Huang, L.-J. Deng, Z.-C. Wu, X. Wu, and G. Vivone, "Variational pansharpening based on coefficient estimation with nonlocal regression," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–15, 2023, Art. no. 5406115, doi: 10.1109/TGRS.2023.3305296.

[13] Z.-C. Wu, T.-Z. Huang, L.-J. Deng, J. Huang, J. Chanussot, and G. Vivone, "LRTCFPan: Low-rank tensor completion based framework for pansharpening," *IEEE Trans. Image Process.*, vol. 32, pp. 1640–1655, 2023.

[14] G. Masi, D. Cozzolino, L. Verdoliva, and G. Scarpa, "Pansharpening by convolutional neural networks," *Remote Sens.*, vol. 8, no. 7, p. 594, 2016.

[15] C. Dong, C. C. Loy, K. He, and X. Tang, "Image super-resolution using deep convolutional networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 38, no. 2, pp. 295–307, Feb. 2015.

[16] J. Yang, X. Fu, Y. Hu, Y. Huang, X. Ding, and J. Paisley, "PanNet: A deep network architecture for pan-sharpening," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 5449–5457.

[17] C. Jin, L.-J. Deng, T.-Z. Huang, and G. Vivone, "Laplacian pyramid networks: A new approach for multispectral pansharpening," *Inf. Fusion*, vol. 78, pp. 158–170, Feb. 2022.

[18] R. Ran, L.-J. Deng, T.-X. Jiang, J.-F. Hu, J. Chanussot, and G. Vivone, "GuidedNet: A general CNN fusion framework via high-resolution guidance for hyperspectral image super-resolution," *IEEE Trans. Cybern.*, vol. 53, no. 7, pp. 4148–4161, Jul. 2023, doi: 10.1109/TCYB.2023.3238200.

[19] Y. Ye, T. Wang, F. Fang, and G. Zhang, "MSCSCformer: Multiscale convolutional sparse coding-based transformer for pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–12, 2024, Art. no. 5405112, doi: 10.1109/TGRS.2024.3391355.

[20] Z. Jin, T. Zhang, T. Jiang, G. Vivone, and L. Deng, "LAGConv: Local-context adaptive convolution kernels with global harmonic bias for pansharpening," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2022, pp. 1113–1121.

[21] H. Lu et al., "AWFLN: An adaptive weighted feature learning network for pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–15, 2023, Art. no. 5400815, doi: 10.1109/TGRS.2023.3241643.

[22] M. Zhou et al., "Spatial-frequency domain information integration for pan-sharpening," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 274–291.

[23] J. Hou, Q. Cao, R. Ran, C. Liu, J. Li, and L.-J. Deng, "Bidomain modeling paradigm for pansharpening," in *Proc. 31st ACM Int. Conf. Multimedia*, Oct. 2023, pp. 347–357.

[24] M. Zhou et al., "A general spatial-frequency learning framework for multimodal image fusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, early access, Feb. 21, 2024, doi: 10.1109/TPAMI.2024.3368112.

[25] X. Meng, N. Wang, F. Shao, and S. Li, "Vision transformer for pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2022, Art. no. 5409011, doi: 10.1109/TGRS.2022.3168465.

[26] X. Su, J. Li, and Z. Hua, "Transformer-based regression network for pansharpening remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–23, 2022, Art. no. 5407423, doi: 10.1109/TGRS.2022.3152425.

[27] W. G. C. Bandara and V. M. Patel, "HyperTransformer: A textural and spectral feature fusion transformer for pansharpening," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Sep. 2022, pp. 1767–1777.

[28] M. Zhou, J. Huang, Y. Fang, X. Fu, and A. Liu, "Pan-sharpening with customized transformer and invertible neural network," in *Proc. AAAI Conf. Artif. Intell.*, vol. 36, 2022, pp. 3553–3561.

[29] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[30] J. Choi, K. Yu, and Y. Kim, "A new adaptive component-substitution-based satellite image fusion by using partial replacement," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 1, pp. 295–309, Jan. 2011.

[31] W. J. Carper, T. M. Lillesand, and R. W. Kiefer, "The use of intensity-hue-saturation transformations for merging SPOT panchromatic and multispectral image data," *Photogramm. Eng. Remote Sens.*, vol. 56, no. 4, pp. 459–467, Apr. 1990.

[32] L. Alparone, A. Garzelli, and G. Vivone, "Intersensor statistical matching for pansharpening: Theoretical issues and practical solutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4682–4695, Aug. 2017.

[33] X. Otazu, M. González-Audícana, O. Fors, and J. Núñez, "Introduction of sensor spectral response into image fusion methods. Application to wavelet-based methods," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 10, pp. 2376–2385, Oct. 2005.

[34] X. Fu, Z. Lin, Y. Huang, and X. Ding, "A variational pan-sharpening with local gradient constraints," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 10257–10266.

[35] K. Zhang, Z. Li, F. Zhang, W. Wan, and J. Sun, "Pan-sharpening based on transformer with redundancy reduction," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[36] X. Wang, X. Wang, R. Song, X. Zhao, and K. Zhao, "MCT-Net: Multi-hierarchical cross transformer for hyperspectral and multispectral image fusion," *Knowl.-Based Syst.*, vol. 264, Mar. 2023, Art. no. 110362.

[37] Q. Wang, X. Jin, Q. Jiang, L. Wu, Y. Zhang, and W. Zhou, "DBCT-Net: A dual branch hybrid CNN-transformer network for remote sensing image fusion," *Expert Syst. Appl.*, vol. 233, Dec. 2023, Art. no. 120829.

[38] S. Venkataramanan, A. H. Ghodrati, Y. Asano, F. Porikli, and A. Habibian, "Skip-attention: Improving vision transformers by paying less attention," in *Proc. 12th Int. Conf. Learn. Represent.*, Jan. 2023, pp. 1–13. [Online]. Available: https://openreview.net/forum?id=vI95kcLAoU

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[39] J. Hou et al., "Linearly-evolved transformer for pan-sharpening," 2024, *arXiv:2404.12804*.
[40] R. H. Yuhas, A. Goetz, and J. Boardman, "Discrimination among semi-arid landscape endmembers using the spectral angle mapper (SAM) algorithm," in *Proc. JPL, Summaries 3rd Annu. JPL Airborne Geosci. Workshop AVIRIS Workshop*, vol. 1, Jun. 1992, pp. 147–149.
[41] L. Wald, *Data Fusion: Definitions Architectures: Fusion Images Different Spatial Resolutions*. Paris, France: Presses des MINES, 2002.
[42] J. Zhou, D. L. Civco, and J. A. Silander, "A wavelet transform method to merge Landsat TM and SPOT panchromatic data," *Int. J. Remote Sens.*, vol. 19, no. 4, pp. 743–757, 1998.
[43] A. Garzelli and F. Nencini, "Hypercomplex quality assessment of multi/hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 662–665, Oct. 2009.
[44] G. Vivone et al., "A critical comparison among pansharpening algorithms," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 5, pp. 2565–2586, May 2014.
[45] L.-J. Deng, G. Vivone, C. Jin, and J. Chanussot, "Detail injection-based deep convolutional neural networks for pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6995–7010, Aug. 2021.
[46] S. Xu, J. Zhang, Z. Zhao, K. Sun, J. Liu, and C. Zhang, "Deep gradient projection networks for pan-sharpening," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 1366–1375.
[47] M. Zhou, J. Huang, C.-L. Guo, and C. Li, "Fourmer: An efficient global modeling paradigm for image restoration," in *Proc. Int. Conf. Mach. Learn.*, 2023, pp. 42589–42601.
[48] S. Lolli, L. Alparone, A. Garzelli, and G. Vivone, "Haze correction for contrast-based multispectral pansharpening," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2255–2259, Dec. 2017.
[49] G. Vivone, "Robust band-dependent spatial-detail approaches for panchromatic sharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6421–6433, Sep. 2019.



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